ASSIGNMENT – VII

TITLE :
Create a Docker container for your AI model and test it on your local machine to ensure consistent performance across environments.

NAME : Shinde Shubham Dnyandev. **ROLL NO :** 23107121. **BATCH :** B

```
admin@ubantu-Precision-3660: ~/ai-docker-app/model
admin@ubantu-Precision-3660:~$ mkdir ai-docker-app
admin@ubantu-Precision-3660:~$ cd ai-docker-app
admin@ubantu-Precision-3660:~/ai-docker-app$ mkdir model
admin@ubantu-Precision-3660:~/ai-docker-app$ cd model
admin@ubantu-Precision-3660:~/ai-docker-app/model$ nano app.py
```

```
admin@ubantu-Precision-3660: ~/ai-docker-app/model
  GNU nano 7.2                                                    app.py *
from flask import Flask
from pymongo import MongoClient

app = Flask(__name__)

client = MongoClient("mongodb://mongodb:27017/")
db = client.testdb

@app.route("/")
def home():
    return "AI Model Running"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
admin@ubantu-Precision-3660:~/ai-docker-app/model$ nano requirements.txt
```

```
admin@ubantu-Precision-3660: ~/ai-docker-app/model
  GNU nano 7.2                                          requirements.txt *
flask
pymongo
```

```
admin@ubantu-Precision-3660:~/ai-docker-app/model$ nano Dockerfile
```

```
admin@ubantu-Precision-3660: ~/ai-docker-app/model
  GNU nano 7.2                                            Dockerfile *
FROM python:3.9

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY app.py .

CMD ["python", "app.py"]
```

```
admin@ubantu-Precision-3660:~/ai-docker-app/model$ cd ..
admin@ubantu-Precision-3660:~/ai-docker-app$ mkdir frontend
admin@ubantu-Precision-3660:~/ai-docker-app$ cd frontend
admin@ubantu-Precision-3660:~/ai-docker-app/frontend$ nano index.html
```

admin@ubantu-Precision-3660: ~/ai-docker-app/frontend

```
  GNU nano 7.2                                                    index.html *
<!DOCTYPE html>
<html>
<head>
        <title>AI Docker App</title>
</head>
<body>
        <h1>AI Docker Application</h1>
        <p>Frontend running in Nginx container.</p>
</body>
</html>
```

```
admin@ubantu-Precision-3660:~/ai-docker-app/frontend$ nano Dockerfile
```

admin@ubantu-Precision-3660: ~/ai-docker-app/frontend

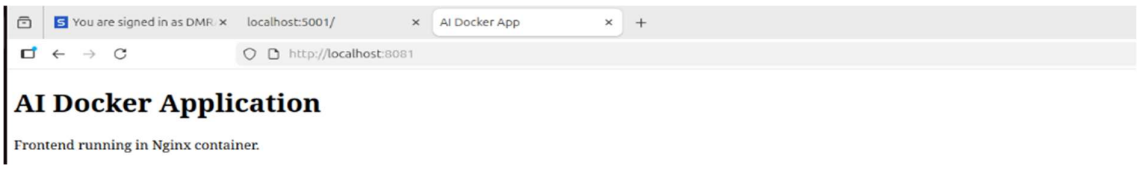
```
  GNU nano 7.2                                                    Dockerfile *
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
```

```
admin@ubantu-Precision-3660:~/ai-docker-app/frontend$ cd ..
admin@ubantu-Precision-3660:~/ai-docker-app$ nano docker-compose.yml
```

admin@ubantu-Precision-3660: ~/ai-docker-app

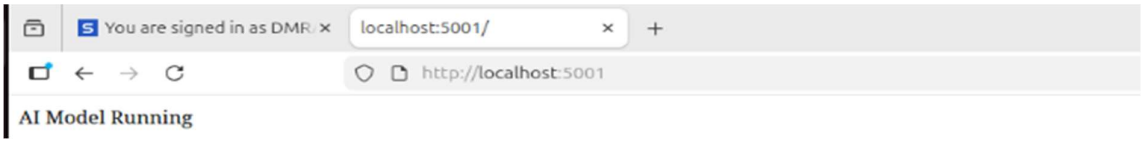
```
  GNU nano 7.2                                                    docker-compose.yml *
version: "3"

services:

  mongodb:
    image: mongo
    container_name: mongo_db
    ports:
      - "27018:27017"

  model:
    build: ./model
    ports:
      - "5001:5000"
    depends_on:
      - mongodb

  frontend:
    build: ./frontend
    ports:
      - "8081:80"
    depends_on:
      - model
```

```
admin@ubantu-Precision-3660: ~/ai-docker-app
admin@ubantu-Precision-3660:~/ai-docker-app$ ls -R
.:
docker-compose.yml  frontend  model

./frontend:
Dockerfile  index.html

./model:
app.py  Dockerfile  requirements.txt
admin@ubantu-Precision-3660:~/ai-docker-app$ sudo docker compose up --build
WARN[0000] /home/admin/ai-docker-app/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
Compose can now delegate builds to bake for better performance.
 To do so, set COMPOSE_BAKE=true.
[+] Building 34.4s (19/19) FINISHED                                                                                    docker:default
 => [model internal] load build definition from Dockerfile                                                                        0.0s
 => => transferring dockerfile: 172B                                                                                              0.0s
 => [model internal] load metadata for docker.io/library/python:3.9                                                               0.9s
 => [model internal] load .dockerignore                                                                                           0.0s
 => => transferring context: 2B                                                                                                   0.0s
 => [model 1/5] FROM docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb3304d07c913          0.0s
 => [model internal] load build context                                                                                           0.0s
 => => transferring context: 368B                                                                                                 0.0s
 => CACHED [model 2/5] WORKDIR /app                                                                                               0.0s
 => [model 3/5] COPY requirements.txt .                                                                                           0.0s
 => [model 4/5] RUN pip install -r requirements.txt                                                                              28.0s
 => [model 5/5] COPY app.py .                                                                                                     0.0s
 => [model] exporting to image                                                                                                    0.3s
 => => exporting layers                                                                                                           0.3s
 => => writing image sha256:72b751b8dcc718922d342ebb770fd71fd8baf117a4fd5c734d91ea167670eeef                                      0.0s
 => => naming to docker.io/library/ai-docker-app-model                                                                            0.0s
 => [model] resolving provenance for metadata file                                                                                0.0s
 => [frontend internal] load build definition from Dockerfile                                                                     0.0s
 => => transferring dockerfile: 104B                                                                                              0.0s
 => [frontend internal] load metadata for docker.io/library/nginx:alpine                                                          2.2s
 => [frontend internal] load .dockerignore                                                                                        0.0s
 => => transferring context: 2B                                                                                                   0.0s
 => [frontend internal] load build context                                                                                        0.0s
 => => transferring context: 207B                                                                                                 0.0s
 => [frontend 1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba      2.8s
 => => resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba              0.0s
 => => sha256:1d13701a5f9f3fb01aaa88cef2344d65b6b5bf6b7d9fa4cf0dca557a8d7702ba 10.33kB / 10.33kB                                  0.0s
 => => sha256:c032460d1fd739783174 79ba23c37bcb57d93156cab122eb3c54b8e4bdc292fa 2.50kB / 2.50kB                                   0.0s
 => => sha256:b76de378d57272a1dd9091a05dd548a3639dfb792ebdbf95d06704d2950afdea 11.65kB / 11.65kB                                  0.0s
 => => sha256:bca5d04786e112d958f100a66f8257b2aeefc14b64d81e405c3c44acff2fb000 1.86MB / 1.86MB                                    0.9s
 => => sha256:3e2c181db1b0985ce357c7aaf48ac615f30f392cd15d5c5ba34c4faa1f4f39a2 626B / 626B                                        0.9s
```



AI Docker Application

Frontend running in Nginx container.



AI Model Running

```
admin@ubantu-Precision-3660: ~/ai-docker-app
admin@ubantu-Precision-3660: ~/ai-docker-app   admin@ubantu-Precision-3660: ~/ai-docker-app
admin@ubantu-Precision-3660:~/ai-docker-app$ sudo docker exec -it mongo_db mongosh
[sudo] password for admin:
Current Mongosh Log ID: 69ae6ec8abc9ae63238563b0
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/


To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

------
   The server generated these startup warnings when booting
   2026-03-09T06:47:39.473+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
   2026-03-09T06:47:39.532+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
   2026-03-09T06:47:39.532+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
   2026-03-09T06:47:39.532+00:00: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile
   2026-03-09T06:47:39.532+00:00: We suggest setting the contents of sysfsFile to 0.
   2026-03-09T06:47:39.532+00:00: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.
------

test> show dbs
admin   40.00 KiB
config  12.00 KiB
local   40.00 KiB
test> use testdb
switched to db testdb
testdb> db.users.insertOne({name:"testuser"})
{
  acknowledged: true,
  insertedId: ObjectId('69ae6ee9abc9ae63238563b1')
}
testdb> db.users.find()
[ { _id: ObjectId('69ae6ee9abc9ae63238563b1'), name: 'testuser' } ]
testdb> exit
admin@ubantu-Precision-3660:~/ai-docker-app$
```

```
admin@ubantu-Precision-3660:~/ai-docker-app$ sudo docker logs ai-docker-app-frontend-1
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/09 06:47:39 [notice] 1#1: using the "epoll" event method
2026/03/09 06:47:39 [notice] 1#1: nginx/1.29.5
2026/03/09 06:47:39 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/03/09 06:47:39 [notice] 1#1: OS: Linux 6.17.0-14-generic
2026/03/09 06:47:39 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/09 06:47:39 [notice] 1#1: start worker processes
2026/03/09 06:47:39 [notice] 1#1: start worker process 30
2026/03/09 06:47:39 [notice] 1#1: start worker process 31
2026/03/09 06:47:39 [notice] 1#1: start worker process 32
2026/03/09 06:47:39 [notice] 1#1: start worker process 33
2026/03/09 06:47:39 [notice] 1#1: start worker process 34
2026/03/09 06:47:39 [notice] 1#1: start worker process 35
2026/03/09 06:47:39 [notice] 1#1: start worker process 36
2026/03/09 06:47:39 [notice] 1#1: start worker process 37
2026/03/09 06:47:39 [notice] 1#1: start worker process 38
2026/03/09 06:47:39 [notice] 1#1: start worker process 39
2026/03/09 06:47:39 [notice] 1#1: start worker process 40
2026/03/09 06:47:39 [notice] 1#1: start worker process 41
2026/03/09 06:47:39 [notice] 1#1: start worker process 42
2026/03/09 06:47:39 [notice] 1#1: start worker process 43
2026/03/09 06:47:39 [notice] 1#1: start worker process 44
2026/03/09 06:47:39 [notice] 1#1: start worker process 45
2026/03/09 06:47:39 [notice] 1#1: start worker process 46
2026/03/09 06:47:39 [notice] 1#1: start worker process 47
2026/03/09 06:47:39 [notice] 1#1: start worker process 48
2026/03/09 06:47:39 [notice] 1#1: start worker process 49
172.18.0.1 - - [09/Mar/2026:06:53:21 +0000] "GET / HTTP/1.1" 200 168 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:148.0) Gecko/20100101 Firefox/148.0" "-"
2026/03/09 06:53:21 [error] 30#30: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.18.0.1, server: localhost, request: "GET /favicon.ic
o HTTP/1.1", host: "localhost:8081", referrer: "http://localhost:8081/"
172.18.0.1 - - [09/Mar/2026:06:53:21 +0000] "GET /favicon.ico HTTP/1.1" 404 153 "http://localhost:8081/" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:148.0) Gecko/20100101 Firefox/148.0
" "-"
```

```
admin@ubantu-Precision-3660:~/ai-docker-app$ sudo docker logs ai-docker-app-model-1
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.18.0.3:5000
Press CTRL+C to quit
172.18.0.1 - - [09/Mar/2026 06:52:32] "GET / HTTP/1.1" 200 -
172.18.0.1 - - [09/Mar/2026 06:52:32] "GET /favicon.ico HTTP/1.1" 404 -
admin@ubantu-Precision-3660:~/ai-docker-app$
```

```
admin@ubantu-Precision-3660: ~/ai-docker-app
admin@ubantu-Precision-3660: ~/ai-docker-app        admin@ubantu-Precision-3660: ~/ai-docker-app
172.18.0.1 - - [09/Mar/2026 06:52:32] "GET /favicon.ico HTTP/1.1" 404 -
admin@ubantu-Precision-3660:~/ai-docker-app$ sudo docker logs mongo_db
{"t":{"$date":"2026-03-09T06:47:39.462+00:00"},"s":"I",  "c":"-",        "id":8991200, "ctx":"main","msg":"Shuffling initializers","attr":{"seed":3689570517}}
{"t":{"$date":"2026-03-09T06:47:39.468+00:00"},"s":"I",  "c":"CONTROL",  "id":97374,   "ctx":"main","msg":"Automatically disabling TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify
--sslDisabledProtocols 'TLS1_0'; to force-enable TLS 1.0 specify --sslDisabledProtocols 'none'"}
{"t":{"$date":"2026-03-09T06:47:39.470+00:00"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"main","msg":"Initialized wire specification","attr":{"spec":{"incomingExternalClient":{"mi
nWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWireVersion":0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient":true}}}
{"t":{"$date":"2026-03-09T06:47:39.471+00:00"},"s":"I",  "c":"CONTROL",  "id":5945603, "ctx":"main","msg":"Multi threading initialized"}
{"t":{"$date":"2026-03-09T06:47:39.471+00:00"},"s":"I",  "c":"CONTROL",  "id":4615611, "ctx":"initandlisten","msg":"MongoDB starting","attr":{"pid":1,"port":27017,"dbPath":"/data/db","
architecture":"64-bit","host":"aefec6c88835"}}
{"t":{"$date":"2026-03-09T06:47:39.471+00:00"},"s":"I",  "c":"CONTROL",  "id":23403,   "ctx":"initandlisten","msg":"Build Info","attr":{"buildInfo":{"version":"8.2.5","gitVersion":"a47
1a13094434666c48a1f75451f2efa49f8f5df","openSSLVersion":"OpenSSL 3.0.13 30 Jan 2024","modules":[],"allocator":"tcmalloc-google","environment":{"distmod":"ubuntu2404","distarch":"x86_64
","target_arch":"x86_64"}}}}
{"t":{"$date":"2026-03-09T06:47:39.471+00:00"},"s":"I",  "c":"CONTROL",  "id":51765,   "ctx":"initandlisten","msg":"Operating System","attr":{"os":{"name":"Ubuntu","version":"24.04"}}}
{"t":{"$date":"2026-03-09T06:47:39.471+00:00"},"s":"I",  "c":"CONTROL",  "id":21951,   "ctx":"initandlisten","msg":"Options set by command line","attr":{"options":{"net":{"bindIp":"*"}
}}}
{"t":{"$date":"2026-03-09T06:47:39.471+00:00"},"s":"W",  "c":"NETWORK",  "id":11621101,"ctx":"initandlisten","msg":"Overriding max connections to honor `capMemoryConsumptionForPreAuthB
uffers` settings","attr":{"limit":200674}}
{"t":{"$date":"2026-03-09T06:47:39.471+00:00"},"s":"I",  "c":"NETWORK",  "id":4648601, "ctx":"initandlisten","msg":"Implicit TCP FastOpen unavailable. If TCP FastOpen is required, set
at least one of the related parameters","attr":{"relatedParameters":["tcpFastOpenServer","tcpFastOpenClient","tcpFastOpenQueueSize"]}}
{"t":{"$date":"2026-03-09T06:47:39.473+00:00"},"s":"I",  "c":"STORAGE",  "id":22297,   "ctx":"initandlisten","msg":"Using the XFS filesystem is strongly recommended with the WiredTiger
 storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem","tags":["startupWarnings"]}
{"t":{"$date":"2026-03-09T06:47:39.473+00:00"},"s":"I",  "c":"STORAGE",  "id":22315,   "ctx":"initandlisten","msg":"Opening WiredTiger","attr":{"config":"create,cache_size=7326M,sessio
n_max=33000,eviction=(threads_min=4,threads_max=4),config_base=false,statistics=(fast),log=(enabled=true,remove=true,path=journal,compressor=snappy),builtin_extension_config=(zstd=(com
pression_level=6)),file_manager=(close_idle_time=600,close_scan_interval=10,close_handle_minimum=2000),statistics_log=(wait=0),json_output=(error,message),verbose=[recovery_progress:1,
checkpoint_progress:1,compact_progress:1,live_restore_progress:1,backup:0,checkpoint:0,compact:0,eviction:0,fileops:0,history_store:0,live_restore:0,recovery:0,rts:0,salvage:0,tiered:0
,timestamp:0,transaction:0,verify:0,log:0],prefetch=(available=true,default=false),"}}
{"t":{"$date":"2026-03-09T06:47:39.484+00:00"},"s":"I",  "c":"WTRECOV",  "id":22430,   "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773038859,"ts_usec"
:484626,"thread":"1:0x7e59d7862400","session_name":"wiredtiger_open","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"
opening the WiredTiger library"}}}
{"t":{"$date":"2026-03-09T06:47:39.504+00:00"},"s":"I",  "c":"WTRECOV",  "id":22430,   "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773038859,"ts_usec"
:504682,"thread":"1:0x7e59d7862400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"conne
ction configuration string parsing completed"}}}
{"t":{"$date":"2026-03-09T06:47:39.507+00:00"},"s":"I",  "c":"WTRECOV",  "id":22430,   "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773038859,"ts_usec"
:507323,"thread":"1:0x7e59d7862400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"start
ing WiredTiger utility threads"}}}
{"t":{"$date":"2026-03-09T06:47:39.508+00:00"},"s":"I",  "c":"WTRECOV",  "id":22430,   "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773038859,"ts_usec"
:508812,"thread":"1:0x7e59d7862400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"start
ing WiredTiger recovery"}}}
{"t":{"$date":"2026-03-09T06:47:39.509+00:00"},"s":"I",  "c":"WTRECOV",  "id":22430,   "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773038859,"ts_usec"
:508984,"thread":"1:0x7e59d7862400","session_name":"txn-recover","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"scan
ning metadata to find the largest file ID"}}}
{"t":{"$date":"2026-03-09T06:47:39.509+00:00"},"s":"I",  "c":"WTRECOV",  "id":22430,   "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773038859,"ts_usec"
```

```
admin@ubantu-Precision-3660:~/ai-docker-app$ sudo docker compose down
WARN[0000] /home/admin/ai-docker-app/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 4/4
 ✔ Container ai-docker-app-frontend-1  Removed                                                    0.2s
 ✔ Container ai-docker-app-model-1     Removed                                                   10.2s
 ✔ Container mongo_db                  Removed                                                    0.2s
 ✔ Network ai-docker-app_default       Removed                                                    0.2s
admin@ubantu-Precision-3660:~/ai-docker-app$
```